

# ZeroDrift Mk3 — buy checklist, one item per row

Total ₹7,059 · ✓ = price verified against a live listing, ~ = estimate · generated from docs/data/bom\_tier\_a.json

**Tick the box for wherever you actually buy each item** — a filled box is my routing guess, an empty one means I'd default to Amar Robotics unless it's clearly a specialist part. I never saw Amar Robotics' actual stock (Instagram/IndiaMart are both blocked from where I work), so treat every box as a suggestion to confirm, not a fact. The five **online** items are the ones with real delivery lead time — order those first, tonight, regardless of what else is still open.

| QTY         | ITEM                                                       |   | ₹      | AMAR<br>ROBOTICS | BLINKIT | ONLINE | HAVE |
|-------------|------------------------------------------------------------|---|--------|------------------|---------|--------|------|
| 2           | NEMA17 stepper motor, 1.8°/step, 4.2 kg-cm                 | ✓ | ₹1,498 | ✓                |         |        |      |
| 2           | A4988 stepper driver module                                | ✓ | ₹340   | ✓                |         |        |      |
| 2           | NEMA17 L-bracket (pan + tilt)                              | ✓ | ₹254   | ✓                |         |        |      |
| 2           | 100 µF 25 V electrolytic capacitor                         | ~ | ₹20    | ✓                |         |        |      |
| 2           | AS5600 magnetic encoder module                             | ✓ | ₹498   |                  |         | ✓      |      |
| 2           | Diametric magnet, 6 × 2.5 mm                               | ~ | ₹180   |                  |         | ✓      |      |
| 1           | TCA9548A I²C multiplexer                                   | ✓ | ₹199   |                  |         | ✓      |      |
| 1           | USB camera module, M12 mount, 12 mm lens                   | ~ | ₹1,400 |                  |         | ✓      |      |
| 1           | Red filter — gel sheet or red acrylic offcut               | ~ | ₹100   |                  |         | ✓      |      |
| —           | KY-008 laser module                                        | ~ | ₹0     | ✓                |         |        |      |
| 1           | ABS project box, head enclosure                            | ~ | ₹115   | ✓                |         |        |      |
| —           | Arduino Nano — tracker                                     | ~ | ₹0     | ✓                |         |        |      |
| 1           | Arduino Nano — beacon                                      | ~ | ₹225   | ✓                |         |        |      |
| 1           | 12 V 3000 mAh Li-ion battery pack + charger, DC barrel out | ~ | ₹650   |                  |         | ✓      |      |
| 1           | DC barrel jack to screw-terminal adapter                   | ~ | ₹30    | ✓                |         |        |      |
| 1           | SPST toggle switch — motor rail kill switch                | ~ | ₹25    | ✓                |         |        |      |
| 1           | Breadboard, 830 point (drivers + Nano + mux)               | ~ | ₹90    | ✓                |         |        |      |
| 2           | USB-B and mini-USB cables for the two Nanos                | ~ | ₹198   |                  | ✓       |        |      |
| 1           | Acrylic sheet 3 mm, 6"×6" (pack of 2)                      | ✓ | ₹199   | ✓                |         |        |      |
| 1           | Vibration motor, 3 V DC                                    | ✓ | ₹20    | ✓                |         |        |      |
| 1           | Transistor + diode + resistor combo kit                    | ✓ | ₹105   | ✓                |         |        |      |
| 1           | High-brightness RED LED, 10 mm, 650 nm                     | ~ | ₹22    | ✓                |         |        |      |
| 1           | 2N2222 transistor + 220 Ω + 1 kΩ                           | ~ | ₹20    | ✓                |         |        |      |
| 1           | Ping-pong ball (diffuser)                                  | ✓ | ₹129   |                  | ✓       |        |      |
| —           | USB power bank (any, 5V) — beacon supply                   | ~ | ₹0     | ✓                |         |        |      |
| 1           | SPST toggle switch                                         | ~ | ₹25    | ✓                |         |        |      |
| 1           | ABS project box                                            | ~ | ₹115   | ✓                |         |        |      |
| 1           | High-brightness WHITE LED (brighter than beacon)           | ~ | ₹22    | ✓                |         |        |      |
| 1           | 68 Ω resistor                                              | ~ | ₹5     | ✓                |         |        |      |
| 1           | 3×AA battery holder + 3 AA cells — decoy supply            | ~ | ₹90    | ✓                |         |        |      |
| 1           | Switch + scrap housing                                     | ~ | ₹35    | ✓                |         |        |      |
| 1           | Dupont jumper wires — M-M, M-F, F-F                        | ~ | ₹150   | ✓                |         |        |      |
| 1           | M3 screws, nuts, standoffs assortment                      | ~ | ₹150   | ✓                |         |        |      |
| 1           | Heat-shrink, solder, hot-glue sticks                       | ~ | ₹150   | ✓                |         |        |      |
| GRAND TOTAL |                                                            |   | ₹7,059 | ₹3,705           | ₹327    | ₹3,027 | ₹0   |

ZeroDrift · Team SIH26169 · ISRO · full guide in docs/submission/ZeroDrift\_Mk3\_Build\_Guide.pdf · checklist generated from the same source file the guide uses, so the two cannot disagree

TOOLS — NOT PART OF THE ₹7,059, NOT THE BUILD

Not part of the build and not in the total. These are what you need in your hand to assemble it. Tick 'have' for anything the team or the college lab already owns -- most teams own most of this. The multimeter is the one that is genuinely not optional.

| TOOL                         | WHY YOU NEED IT                                                                                                                                                                                                               | ₹ IF BUYING | HAVE | BUY |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|-----|
| Multimeter                   | Setting the A4988 Vref is a voltage measurement on the driver's trim pot. Without it you are guessing the motor current: too low and the gimbal skips steps, too high and the motor cooks. There is no way to do this by eye. | ₹500        |      |     |
| Small flat screwdriver, 2 mm | The A4988 trim pot is tiny and a normal screwdriver slips off it onto the board.                                                                                                                                              | ₹150        |      |     |
| Wire stripper / side cutter  | Thirty-odd connections.                                                                                                                                                                                                       | ₹150        |      |     |
| Soldering iron, 25-40 W      | The beacon and decoy are soldered; solder itself is already in consumables.                                                                                                                                                   | ₹350        |      |     |
| Hot glue gun                 | Strain relief on every connector that leaves a moving axis. Glue sticks are already in consumables.                                                                                                                           | ₹250        |      |     |
| Allen key set (M3)           | NEMA17 brackets use M3 socket screws.                                                                                                                                                                                         | ₹120        |      |     |
| Zip ties + double-sided tape | Cable management on a gimbal is not cosmetic: a cable that snags is a lost step and a lost lock.                                                                                                                              | ₹80         |      |     |
| USB extension cable, 1.5 m   | The camera rides on the gimbal; its cable has to reach the laptop with slack for the full pan range.                                                                                                                          | ₹150        |      |     |
| IF YOU OWN NONE OF IT        |                                                                                                                                                                                                                               | ₹1,750      |      |     |